Fiscal Year:	FY 2024	**Task Last Updated:**	FY 03/15/2024
PI Name:	Schreckenghost, Debra M.E.E.		
Project Title:	HCAAM VNSCOR: Enhancing Situation Awareness of Automated Procedures Using Adaptive Multimodal Augmented Reality Displays		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **HSIA**:Risk of Adverse Outcomes Due to Inadequate Human Systems Integration Architecture		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	ghost@ieee.org	**Fax:**	FY
PI Organization Type:	INDUSTRY	**Phone:**	281-461-7886
Organization Name:	TRACLabs, Inc.		
PI Address 1:	1331 Gemini Street		
PI Address 2:	Suite 100		
PI Web Page:			
City:	Webster	**State:**	TX
Zip Code:	77058	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2017-2018 HERO 80JSC017N0001-BPBA Topics in Biological, Physiological, and Behavioral Adaptations to Spaceflight. Appendix C
Start Date:	05/15/2019	**End Date:**	03/31/2025
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
Contact Email:	alexandra.m.whitmire@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 03/31/2025 per A. Beitman/JSC (Ed., 9/19/24) NOTE: End date changed to 12/31/2024 per A. Beitman/JSC (Ed., 1/20/23) NOTE: End date changed to 5/14/2023 per S. Huppman/HRP and NSSC information (Ed., 3/3/2020)		
Key Personnel Changes/Previous PI:	March 2024 report: There are no key personnel changes.		
COI Name (Institution):	Holden, Kritina Ph.D. (NASA Johnson Space Center) Dory, Jonathan B.S. (NASA Johnson Space Center)		
Grant/Contract No.:	80NSSC19K0667		
Performance Goal No.:			
Performance Goal Text:			

Task Description:

This task is part of the Human Capabilities Assessments for Autonomous Missions (HCAAM) Virtual NASA Specialized Center of Research (VNSCOR).
Future deep space missions will present new challenges for crew, and increased risks to human performance due to the stress, fatigue, radiation exposure, and isolation that characterizes these missions. In addition, crew will no longer be able to depend on timely support from Mission Control due to distance from the Earth, but will have to work autonomously, while maintaining high performance. Mission Controllers may not be available to answer questions, check system status, assist with procedures, monitor for errors, or troubleshoot problems. Greater crew autonomy will increase dependence on automated systems, and design of these automated systems must be driven by sound human-system integration standards and guidelines in order to ensure mission success. Historically, crew have had very limited dependence on automated systems, thus crew will be faced with a new way of working that may put situation awareness (SA) at risk. We must develop methods for promoting good situation awareness in the automated systems that will most certainly be part of future deep space vehicles and habitats.

Procedure automation is a promising technology for reducing crew workload. We define procedure automation as technology that automates the selection or execution of procedural tasks. Structuring the work of automation according to human procedures should improve the transparency of automation actions. This approach provides a means for establishing common ground about ongoing tasks to improve operator understanding of automation behavior.

New technologies such as adaptive, multimodal, augmented reality displays can offer the benefits of information presentation tailored to meet the needs of each crewmember, taking into consideration the current state of that crewmember (e.g., sleep-deprived, high workload), as well as the current state of his/her environment and ongoing activities (e.g., emergency situation, time-critical operations).

We propose to combine technology for procedure automation with technology for augmented reality multi-modal (ARMM) user interfaces using Microsoft Hololens head-mounted display to provide a virtual task assistant to assist crew in performing procedural work. This virtual task assistant will be capable of identifying which procedures should be performed, performing actions in crew procedures, and summarizing actions taken by the human-automation team to assist crew in preparing for tasks and taking over tasks from other team members.

Four studies are planned to evaluate the effects of a virtual task assistant combining procedure automation with augmented reality multi-modal (AARM) user interfaces on human task performance. These studies will achieve the following aims:

Aim. Determine best methods to improve situation awareness and improve crew autonomy when using a virtual task assistant to prepare for and perform manual maintenance.

Aim. Determine best practices to improve crew performance when using virtual task assistant to aid in task transitions (task handovers, automation interruption).

Aim. Determine best methods for display of procedural work to improve crew autonomy and satisfaction

The proposed work addresses a number of gaps in the Human Research Program Human Factors and Behavioral Performance risks. This project will provide guidelines for designing effective human-automation systems and evaluate human-automation performance for exemplar procedure automation systems. This project also will provide guidance for the application of multi-modal and adaptive displays and control to Human-Computer Interaction (HCI) design for long duration operations.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Technologies for virtual task assistance are increasingly available in everyday life. One of the most common is voice enabled assistance, like Siri and Alexa, that aid some activities of daily living. And augmented and virtual reality technologies are becoming mainstream, with the introduction of new devices such as Microsoft HoloLens 2 and Apple's Vision Pro.
The Virtual Intelligent Task Assistant (VITA) project is leveraging augmented reality platforms and toolkits for augmented reality apps to develop a virtual task assistant that can be used to assist users with procedural task work on the job. Our technical approach is innovative in that new procedural tasks can be supported without custom software development. Our experimental research is distinguished by investigating effective task assistance for maintenance or assembly tasks where hands-free operation of task assistance is beneficial. For the first study we investigated best techniques for using augmented reality task assistance when assembling small rovers that are held in the hands during assembly. Subsequent studies investigate other aspects of using augmented reality for assembly tasks, including use of 3D images in procedures and virtual notification for task milestones and transitions.

This technology and associated research findings have potential benefit to NASA for the assembly, maintenance, and repair of aircraft, spacecraft, habitats, and robotics. This technology and associated research findings also have broader potential benefit for any organization performing assembly and maintenance procedural work. This includes assembly and maintenance of drilling equipment for the oil and gas industry, equipment used in chemical processing plants, and maintenance and repair of commercial aircraft.

I. VITA Study in HERA Campaign 6
The Virtual Intelligent Task Assistant (VITA) project conducted a study in NASA Human Exploration Research Analog (HERA) Campaign 6 (C6) to determine the best methods to improve situation awareness (SA) and improve crew autonomy when using a virtual task assistant to prepare for and perform manual maintenance and assembly. This study attempts to answer the question posed in the proposal:

"Can the virtual task assistant stand-in for Mission Control Center (MCC) and help crew prepare for and perform manual tasks that are not done frequently, such as equipment maintenance and assembly?"

The VITA study conducted in HERA Campaign 6 started in September 2021 and ended in March 2023. We completed data collection for HERA C6 during the reporting period. Prior to each mission, we trained the crew on how to interact with the VITA. After each mission, we debriefed each crew member about their experience using VITA during the mission. Data collection and analysis for the VITA study in HERA C6 was performed during the reporting period. Lori

Task Progress:

Chappell, a NASA biostatistician, performed a statistical analysis of the VITA data collected during HERA Campaign 6.

The findings from all four missions, with 16 participants, are reported in the VITA annual report and in a paper to be presented at International Conference on Environmental Systems (ICES) 2024 (Schreckenghost et al., 2024). These findings are summarized below.

The VITA study investigates the use of techniques to aid crewmembers with procedural tasks when MCC cannot assist. A key aspect of that assistance is to prompt user actions during procedural tasks when MCC is not available in real-time. Three approaches were considered – HoloLens, Tablet, and Team. This study demonstrates that on average a single crewmember using either the Tablet or HoloLens system can complete assembly tasks with fewer total crew hours than required when a second crew member replaces MCC support, as needed for the Team condition.

This study is unique from earlier studies of augmented reality for procedural work displays (Furuya et al., 2018; Markov-Vetter et al., 2013). We use a modified Subjective Awareness Rating Technique (SART) (Taylor, 1990) for collecting SA data during procedure execution. We found only small differences in SA across all conditions, with one exception. Task demand in SART for HoloLens was on average one point higher than the other conditions. We believe this difference is due to increased demand to resolve the technology reliability issues described below.

Not all the expected benefits of hands-free operation were observed with the HoloLens, however. Rover assembly using the HoloLens was expected to take less schedule time and to reduce workload over other conditions. Instead, workload and schedule time were higher for the HoloLens than for both Tablet and Team. A key contributor to this increased workload and schedule time was some reliability and usability issues with the HoloLens system, as evidenced by the lower usability assessment values for this system. Another contributor to reduced usability and increased workload for some crew members was insufficient practice with the HoloLens.

We addressed these performance issues in the VITA study in HERA Campaign 7. Modifications include updating from the HoloLens 1 to the HoloLens 2 virtual reality headset, investigating voice interaction with VITA, and increasing the time users have to familiarize themselves with the HoloLens system.

II. VITA Study in HERA Campaign 7

The aim of the VITA study in HERA Campaign 7 (C7) is to determine the best methods for the display of procedural work to improve crew autonomy and satisfaction. Specifically, this study compares crew performance and satisfaction on rover assembly tasks similar to those used in HERA Campaign 6 but with procedures that vary the level of detail in text instruction and the style of figures, to improve support for flexible execution of actions. These differences are illustrated in existing VITA procedures for rover assembly and disassembly.

The VITA task assistant from HERA C6 was modified for use in HERA C7. A number of changes were made to the VITA task assistant software. The VITA task assistant now runs on the HoloLens 2, instead of the HoloLens 1 used in Campaign 6. The VITA augmented reality display was redesigned, including changing from gaze interaction used in HERA C6 to voice interaction for HERA C7.

To address our aim of best methods for the display of procedure work, we investigate procedures that vary the style of figure and the level of detail in the text. Specifically, we look at the style of the figure, comparing performance with 2D photos to 3D figures with diagrams. We also look at the level of detail in text instructions, comparing procedures with and without more detailed text instructions.

One of the findings from HERA C6 was the need to improve crew familiarity with augmented reality technology. For HERA C7 the time using VITA has been increased by adding more practice with the system. Each crew member uses the VITA AR system with two training procedures during pre-mission training. Then each crew member uses the VITA AR system for six of eight sessions during the mission. For HERA C7 all assembly procedures are performed twice, while for HERA C6 procedures were never repeated. The time between repetitions is counterbalanced between 1 week and 3 weeks.

The VITA study in HERA Campaign 7 will have a total of 16 participants. Participants are selected by HERA to be astronaut-like. Participants are trained prior to HERA and sessions are counterbalanced. Data collection includes situation awareness, workload, usability, and task timing at the end of each session. A survey is administered at the end of each session (called the end of condition survey) and after the final VITA session (called the end of mission survey). These surveys will include questions about user satisfaction and perception of autonomy when using different procedure styles will be added to the end-of-mission survey.

During the reporting period, the VITA project met all HERA C7 delivery milestones. This includes delivering modified rover kits, finishing software upgrades to the VITA task assistant, training the crew of C7M1, and supporting VITA sessions as needed during C7M1.

III. Laboratory Study of VITA Support for Task Transitions

A laboratory study will be performed investigating aid from the VITA task assistant for task transitions. The aim of this study is to determine the best practices to improve crew performance when using the VITA virtual task assistant to aid in task transitions, such as handling the interruption of a manual task by an automated task. In this study, the user is responsible for manually assembling one of the rover configurations. Concurrently they are responsible for supervising an automated task (e.g., the automated startup of a habitat system). Supervisory tasks include monitoring the progress of the startup and intervening when automation pauses for user interventions.

The participant monitors progress on automated tasks using virtual notices displayed in the VITA task assistant. These notices are issued when task milestones are achieved, corresponding to changes in system states or environmental levels.

The VITA virtual notification software design groups notices into panes associated with key habitat systems. Notice panes are shown in the periphery of the field of view (FOV), but can be moved to a different location. Notice panes do not follow head motion.

Notices also are issued in the VITA task assistant when automation pauses, waiting for the supervisor to intervene. The participant intervenes when configuration information is needed or an automated task fails.

A request for user intervention is considered a high saliency notice. The user’s attention should be shifted to high saliency notices as soon as they are received. Each high saliency notice is displayed in a separate pane. The background

	of this pane is different than in the group panes. These notices follow head motion to remain in the user FOV. They must be acknowledged to be moved out of the FOV. This study evaluates two conditions – with and without virtual notices from the automated task displayed in the task assistant. The condition With Virtual Notices will display notices in the HoloLens as described above. The condition Without Virtual Notices will not display notices at all, neither in the HoloLens nor in a Tablet. Instead, the participant will monitor automated task progress and the need for intervention by periodically checking the procedure display on a laptop. Intervention into the automation will be done for both conditions using PRIDE procedures displayed in a browser on a laptop. Task performance will be compared with and without virtual notices in the VITA task assistant. This is a within-subjects study. The augmented reality interface used for the VITA task assistant in this study incorporates the changes made for HERA Campaign 7, such as running as an app on the HoloLens 2. Users interact with procedural task assistance using voice commanding and gestures. The VITA task assistant has been modified to include virtual notification, as described earlier. The VITA laboratory study will target up to 16 participants. The study will be conducted in the Human Factors Engineering Laboratory (HFEL) onsite at NASA Johnson Space Center (JSC) using subjects from the Human Test Subject Facility (HTSF). Participants will be trained prior to the study session and sessions will be counterbalanced. Data collection will include situation awareness, workload, usability, and task timing at the end of each session. A survey will be administered after the VITA session to collect subjective ratings and comments. References 1. Furuya, Hiroshi and Wang, Lui and Elvezio, Carmine and Feiner, Steven. “A Comparative Ground Study of Prototype Augmented Reality Task Guidance for International Space Station Stowage Operations.” 69th International Astronautical Congress. Bremen, Germany, October 1-5, 2018. 2. Markov-Vetter, Daniela and Millberg, Jessica and Staadt, Oliver. “Mobile Augmented Reality for Space Operation Procedures: A Generic Approach of Authoring and Guiding On-Board Payload Activities.” 64th International Astronautical Congress (IAC). Beijing, China, September 23-27, 2013. 3. Schreckenghost D, Holden K, Milam T, Robertson, IWT. “Performance of a Virtual Intelligent Task Assistant for NASA Astronauts Working Remotely during Deep Space Missions.” 53rd International Conference on Environmental Systems (ICES). Louisville, KY, July 21-15, 2024. 4. Taylor RM. Situational awareness rating technique (SART): The development of a tool for aircrew systems design. In: Situational Awareness in Aerospace Operations (AGARD-CP-478). Neuilly Sur Seine, France: NATO - AGARD. p. 3/1–3/17.
Bibliography Type:	Description: (Last Updated: 03/25/2025)
Abstracts for Journals and Proceedings	Schreckenghost D, Holden K, Milam T, Robertson IWT. "Enhancing situation awareness of automated procedures using adaptive multimodal augmented reality displays." 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. Abstracts. 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. , Feb-2024
Papers from Meeting Proceedings	Schreckenghost D, Holden K, Milam T, Robertson IWT. "Performance of a Virtual Intelligent Task Assistant for NASA Astronauts Working Remotely during Deep Space Missions." 53rd International Conference on Environmental Systems (ICES), Louisville, KY, July 21-15, 2024. 53rd International Conference on Environmental Systems (ICES), Louisville, KY, July 21-15, 2024. , Jul-2024